Marginalization keeps the matching blocks

$$\begin{aligned} \boldsymbol{x} &\sim \mathcal{N}(\mu, \Sigma) \\ p(x_1) &= \int p(x_1, x_2)\, \mathrm{d}x_2 \\ &= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11}) \end{aligned}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values

Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

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$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = \begin{bmatrix} I & 0 \end{bmatrix}$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

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- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

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Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

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$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:
zero cross-covariance gives independence

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$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

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- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

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- Conditioning cannot increase variance in any direction

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Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

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From the mean proof: r and x_2 are independent

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- Conditioning cannot increase variance in any direction

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

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- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values